

Pre-Delinquency Intervention Engine: Project Report

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1. Executive Summary

The **Pre-Delinquency Intervention Engine** is a proactive financial risk management system. Unlike traditional models that react to defaults after they happen, this system predicts high-risk behavior *before* a customer misses a payment. It leverages machine learning to score customers daily and empowers risk officers with a real-time dashboard to trigger preventative interventions (e.g., payment plans, courtesy calls).

2. System Architecture

The project follows a modern, event-driven architecture designed for scalability and low latency.

2.1 Core Components

Risk API (FastAPI):

- Serves as the central inference engine.
- Endpoints:
 - GET /customers/high-risk: Retrieves the daily "At-Risk" queue.
 - POST /score: Calculates real-time risk scores using XGBoost.
 - POST /notify: Triggers intervention alerts (simulated SNS/Email).
- **Latency Protection:** Enforces strict SLA checks (default 100ms) for feature retrieval.

Dashboard (Streamlit):

- Front-end interface for Risk Officers.
- Features:
 - **Live Monitor:** View high-risk customers, risk distribution, and key metrics.
 - **Deep Dive:** Detailed customer profile with financial stress trends.
 - **Expandable Explanations:** Uses SHAP values to explain *why* a customer is high risk (e.g., "Utilization spike").
 - **System Logs:** Real-time view of system events (SNS notifications, Kafka streams).

Data Persistence & Simulation:

- **FileBasedCache:** mimicking Redis for persistent storage of customer profiles and feature data.
- **FileBasedSNS:** Simulates AWS SNS for dispatching notifications to log files for auditability.

3. The Machine Learning Model

3.1 Model Choice: XGBoost Classifier

We utilize **XGBoost (Extreme Gradient Boosting)**, a state-of-the-art algorithm for tabular data. It was chosen for its: - **Accuracy:** Superior performance on structured financial data. - **Speed:** Optimized for fast inference, crucial for real-time scoring. - **Explainability:** Compatibility with SHAP (SHapley Additive exPlanations).

3.2 Feature Engineering

The model does not rely on raw data alone. It uses sophisticated behavioral features derived from transaction history: - **Trend Features**: `late_payment_trend` (Is the customer paying later than usual?) - **Variance Features**: `high_balance_variance` (Is their spending erratic?) - **Ratio Features**: `utilization_spike` (Credit limits ratio changes).

3.3 Explainability (SHAP)

A "Black Box" model is unacceptable in finance due to regulatory requirements. We integrated **SHAP** values to provide: - **Global Interpretability**: Which features drive risk across the portfolio? - **Local Interpretability**: Why is *Customer X* specifically high risk? (e.g., +20 points due to "Income Drop", -5 points due to "Long Credit History").

4. Operational Workflow

1. **Ingestion**: Transaction data is ingested daily (simulated).
2. **Scoring**: The Risk API scores all active customers.
3. **Flagging**: Customers with score > 70 are added to the "High Risk" queue.
4. **Review**: A Risk Officer logs into the Dashboard and reviews the queue.
5. **Intervention**: The Officer clicks "Trigger Callback" for a flagged customer.
6. **Action**: The system logs the intervention request to the audit trail (simulated SNS).

5. Future Scope

To evolve this prototype into a production-grade enterprise solution, the following roadmap is proposed:

Phase 4: Advanced MLOps

- **Model Registry (MLflow)**: Version control for model iterations.
- **Feature Store (Feast)**: Replace file-based emulation with a production Feature Store for consistent training/serving skew.
- **Drift Detection**: Automated alerts if the live data distribution diverges from training data.

Phase 5: Cloud Native Deployment

- **Containerization**: Dockerize the API and Dashboard.
- **Orchestration**: Deploy on Kubernetes (EKS/AKS) for auto-scaling.
- **Real Streaming**: Replace simulations with actual AWS Kinesis or Apache Kafka.

Phase 6: Enhanced Analytics

- **A/B Testing**: Test different intervention strategies (e.g., SMS vs. Email) to measure cure rates.
 - **Graph Neural Networks**: Incorporate social network linkages to detect fraud rings.
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Conclusion The Pre-Delinquency Intervention Engine demonstrates a complete end-to-end workflow from raw data to actionable business insight. By catching risk early, it protects the lender's bottom line while offering struggling customers a chance to remediate their status before damaging their credit score.